

IF / ELSE / ELSE IF (REAL-WORLD LOGIC)

1. Write a program to check whether a number is **positive, negative, or zero**.
2. Given marks of a student, decide **Pass or Fail** (pass if marks ≥ 40).
3. Write logic to check whether a number is **even or odd**.
4. Given age, classify as **Child, Teen, Adult, Senior**.
5. Write a program to find the **largest of two numbers**.
6. Write a program to find the **largest of three numbers** using else if.
7. Check whether a year is a **leap year**.
8. Given temperature, decide **Cold, Warm, or Hot**.
9. Check whether a character is a **vowel or consonant**.
10. Given salary, calculate bonus:
 - Salary $\geq 50,000 \rightarrow 20\%$
 - Salary $\geq 30,000 \rightarrow 10\%$
 - Else $\rightarrow 5\%$

SWITCH CASE QUESTIONS

11. Use switch to print the **day of the week** (1–7).
12. Use switch to build a **simple calculator** (+, −, *, /).
13. Use switch to print **month name** from month number.
14. Use switch to check **grade** (A, B, C, D, F).
15. Use switch to display **traffic light actions** (Red, Yellow, Green).

FOR LOOP – BASIC TO LOGIC

16. Print numbers from **1 to 10**.
17. Print numbers from **10 to 1**.
18. Print all **even numbers** from 1 to 50.

Java Script Control Flow & Loops

19. Print all **odd numbers** from 1 to 50.
 20. Find the **sum of first N natural numbers**.
 21. Print the **multiplication table** of a given number.
 22. Count how many numbers between 1–100 are divisible by 3.
 23. Find the **factorial** of a number.
 24. Reverse a number using a loop.
 25. Count the number of digits in a number.
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● WHILE & DO-WHILE LOOP QUESTIONS

26. Print numbers from 1 to 10 using while loop.
 27. Print numbers from 10 to 1 using while loop.
 28. Find the sum of digits of a number using while loop.
 29. Check whether a number is a **palindrome** using while loop.
 30. Print numbers at least once using do...while even if condition is false.
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● FOR...IN & FOR...OF QUESTIONS

31. Iterate over an object and print **keys and values** using for...in.
 32. Iterate over an array and print **all elements** using for...of.
 33. Find the sum of array elements using for...of.
 34. Count how many properties exist in an object.
 35. Explain why for...in should not be used for arrays.
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● BREAK & CONTINUE (INTERVIEW FAVORITES)

36. Print numbers from 1–10 but **stop when number is 5**.
37. Print numbers from 1–10 but **skip number 5**.
38. Find the first number divisible by 7 between 1–100 and stop.

39. Print only odd numbers between 1–20 using continue.
40. Explain difference between break and continue with example.

● PATTERN PROGRAMS (VERY IMPORTANT)

41. Print this pattern:

```
*  
**  
***  
****
```

42. Print this pattern:

```
****  
***  
**  
*
```

43. Print number pattern:

```
1  
12  
123  
1234
```

44. Print number pattern:

```
1  
22  
333  
4444
```

45. Print this pattern:

```
*  
* *  
* * *  
* * * *
```

46. Print square pattern:

```
****  
****  
****  
****
```

47. Print this pattern using nested loops:

```
1 2 3  
1 2 3  
1 2 3
```

THINKING & SCENARIO QUESTIONS

48. Why does an **infinite loop** occur? Give one example.
 49. When should you use while instead of for?
 50. Write a real-world example where do...while is the best choice.
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☒ INSTRUCTIONS

- Solve all 50 questions on paper or editor
- Focus on **logic & flow**, not syntax only

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